

lab report 1 - DSP

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April 2026

Lab 1: Adaptive Filtering

prep exercises set 1 : Adaptive Linear Combiner

a) Block Diagram and Definitions

For a discrete linear FIR filter with $L + 1$ taps, the relationship between the input signal x_k and the output y_k is defined as:

$$y_k = \sum_{l=0}^L w_l x_{k-l} = \mathbf{w}^T \mathbf{x}_k \quad (1)$$

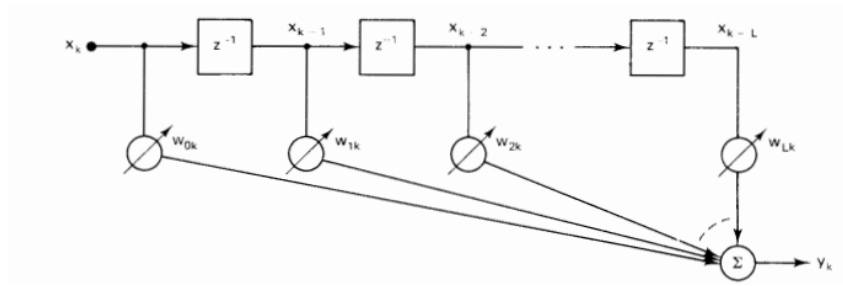


Figure 1: Adaptive FIR Filter Structure.

Where:

- $\mathbf{w} = [w_0, w_1, \dots, w_L]^T$ is the weight vector.
- $\mathbf{x}_k = [x_k, x_{k-1}, \dots, x_{k-L}]^T$ is the input vector at time k .
- z^{-1} denotes a unit delay in the shift register.

b) The Cost Function (MSE)

The error signal ϵ_k is the difference between the desired response d_k and the filter output y_k :

$$\epsilon_k = d_k - y_k = d_k - \mathbf{w}^T \mathbf{x}_k \quad (2)$$

The cost function ξ is defined as the Mean Square Error (MSE):

$$\xi = E[\epsilon_k^2] = E[(d_k - \mathbf{w}^T \mathbf{x}_k)^2] \quad (3)$$

Expanding the quadratic form:

$$\xi = E[d_k^2] + \mathbf{w}^T E[\mathbf{x}_k \mathbf{x}_k^T] \mathbf{w} - 2E[d_k \mathbf{x}_k^T] \mathbf{w} \quad (4)$$

By defining the following statistical parameters:

- $\mathbf{R} = E[\mathbf{x}_k \mathbf{x}_k^T]$ (The Autocorrelation Matrix of the input).
- $\mathbf{P} = E[d_k \mathbf{x}_k]$ (The Cross-Correlation Vector between input and desired).

The MSE is a quadratic function of the weights, forming a hyper-paraboloid:

$$\xi = E[d_k^2] + \mathbf{w}^T \mathbf{R} \mathbf{w} - 2\mathbf{P}^T \mathbf{w} \quad (5)$$

c) Optimal Weight Vector

To find the minimum MSE, we calculate the gradient of ξ with respect to $\mathbf{w}$ and set it to zero:

$$\nabla_{\mathbf{w}} \xi = 2\mathbf{R} \mathbf{w} - 2\mathbf{P} = 0 \quad (6)$$

Solving for the optimal weights $\mathbf{w}^*$:

$$\mathbf{w}^* = \mathbf{R}^{-1} \mathbf{P} \quad (7)$$

It assumes that $\mathbf{R}$ is non-singular.

d) Minimum MSE and Performance

Substituting $\mathbf{w}^*$ back into the cost function gives the minimum error ξ_{min} :

$$\xi_{min} = E[d_k^2] - \mathbf{P}^T \mathbf{R}^{-1} \mathbf{P} = E[d_k^2] - \mathbf{P}^T \mathbf{w}^* \quad (8)$$

To prove this is a unique minimum, we can express the MSE for any arbitrary weight vector $\mathbf{w}$ as:

$$\xi = \xi_{min} + (\mathbf{w} - \mathbf{w}^*)^T \mathbf{R} (\mathbf{w} - \mathbf{w}^*) \quad (9)$$

Since $\mathbf{R}$ is positive-definite, the term $(\mathbf{w} - \mathbf{w}^*)^T \mathbf{R} (\mathbf{w} - \mathbf{w}^*)$ is always positive for $\mathbf{w} \neq \mathbf{w}^*$, confirming that $\mathbf{w}^*$ is

e) Relationship between the MSE and the Optimal Solution

To demonstrate that the performance function ξ is a quadratic surface centered at $\mathbf{w}^*$, we can express any weight vector as a deviation from the optimal:

$$\mathbf{w} = \mathbf{w}^* + \Delta\mathbf{w} \quad (10)$$

Recalling the original performance function:

$$\xi = E[d_k^2] + \mathbf{w}^T \mathbf{R} \mathbf{w} - 2\mathbf{P}^T \mathbf{w} \quad (11)$$

Using the Wiener-Hopf identity $\mathbf{P} = \mathbf{R}\mathbf{w}^*$ and the definition of the minimum error $\xi_{min} = E[d_k^2] - (\mathbf{w}^*)^T \mathbf{R} \mathbf{w}^*$, we substitute $\mathbf{w} = \mathbf{w}^* + \Delta\mathbf{w}$ into the quadratic form:

$$\xi = \xi_{min} + (\mathbf{w} - \mathbf{w}^*)^T \mathbf{R} (\mathbf{w} - \mathbf{w}^*) \quad (12)$$

Derivation Steps:

1. Expand the term $(\mathbf{w} - \mathbf{w}^*)^T \mathbf{R} (\mathbf{w} - \mathbf{w}^*)$:

$$\mathbf{w}^T \mathbf{R} \mathbf{w} - \mathbf{w}^T \mathbf{R} \mathbf{w}^* - (\mathbf{w}^*)^T \mathbf{R} \mathbf{w} + (\mathbf{w}^*)^T \mathbf{R} \mathbf{w}^* \quad (13)$$

2. Since $\mathbf{R}$ is a symmetric matrix, $\mathbf{w}^T \mathbf{R} \mathbf{w}^* = (\mathbf{w}^*)^T \mathbf{R} \mathbf{w}$. Thus:

$$= \mathbf{w}^T \mathbf{R} \mathbf{w} - 2(\mathbf{w}^*)^T \mathbf{R} \mathbf{w} + (\mathbf{w}^*)^T \mathbf{R} \mathbf{w}^* \quad (14)$$

3. Adding $\xi_{min} = E[d_k^2] - (\mathbf{w}^*)^T \mathbf{R} \mathbf{w}^*$ to this result:

$$\xi = E[d_k^2] - (\mathbf{w}^*)^T \mathbf{R} \mathbf{w}^* + \mathbf{w}^T \mathbf{R} \mathbf{w} - 2(\mathbf{w}^*)^T \mathbf{R} \mathbf{w} + (\mathbf{w}^*)^T \mathbf{R} \mathbf{w}^* \quad (15)$$

4. The terms $(\mathbf{w}^*)^T \mathbf{R} \mathbf{w}^*$ cancel out, returning the expression to the original cost function:

$$\xi = E[d_k^2] + \mathbf{w}^T \mathbf{R} \mathbf{w} - 2\mathbf{P}^T \mathbf{w} \quad (16)$$

This confirms that the error surface is a hyper-paraboloid. Since the autocorrelation matrix $\mathbf{R}$ is positive-definite, any deviation $\Delta\mathbf{w}$ from $\mathbf{w}^*$ results in an increase in the MSE, proving that $\mathbf{w}^*$ is the unique global minimum.

f) Solutions to Widrow & Stearns Chapter 2 Exercises

Exercise 6

we will derive the expression for the error from the block diagram from the exercise:

$$\begin{aligned}\varepsilon_k &= d_k - w_1 x_k \\ \zeta &= \mathbb{E}[\varepsilon_k^2] = \mathbb{E}[(d_k - w_1 x_{k-1})^2] \\ \dots &= \mathbb{E}[d_k^2] - 2w_1 \mathbb{E}[d_k x_{k-1}] + w_1^2 \mathbb{E}[(x_k - 1)^2]\end{aligned}$$

$$\begin{aligned}\mathbb{E}[d_k^2] &= 4; \mathbb{E}[d_k x_{k-1}] = 1; \mathbb{E}[d_k x_k] = -1; \mathbb{E}[x_{k-1}^2] = 1 \\ \implies \zeta &= 4 - 2w + w^2 = 3 + (w - 1)^2\end{aligned}$$

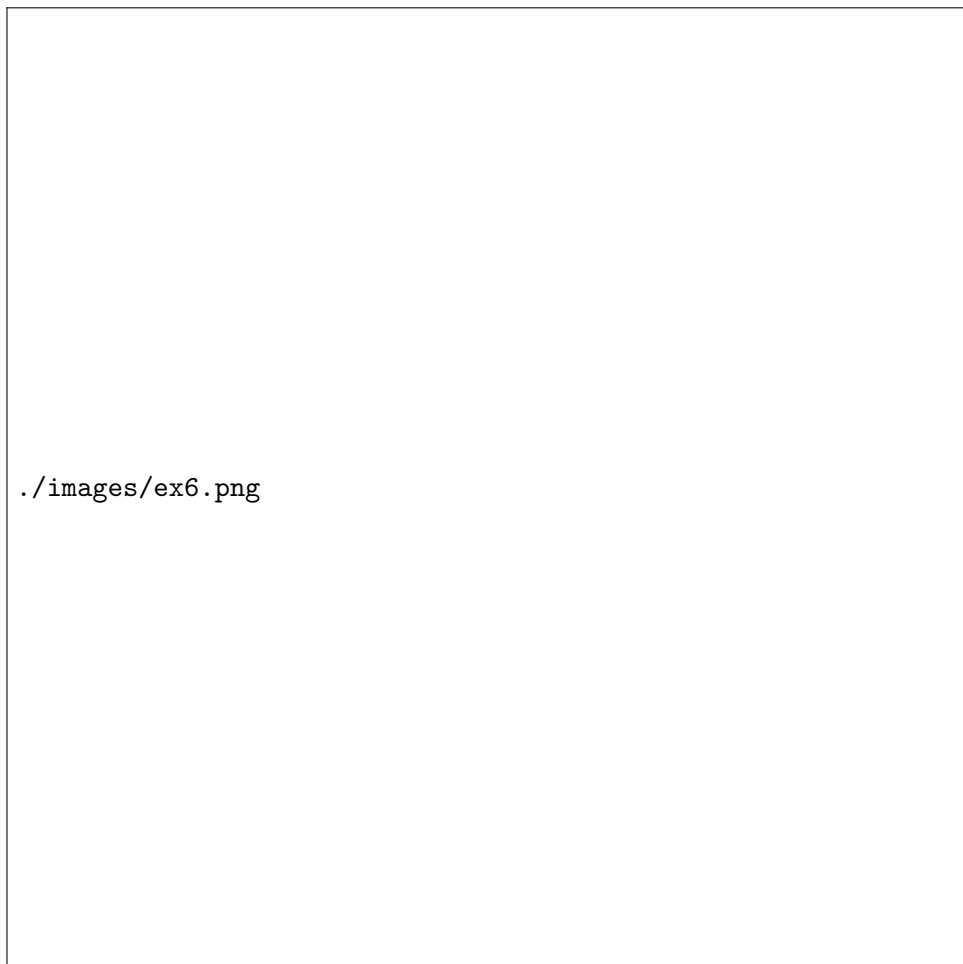


Figure 2: performance function plotted for ex.6

Exercise 7

we will derive the expression for the error from the block diagram from the exercise, this time with the switch s closed:

$$\begin{aligned}\varepsilon_k &= d_k - (x_k + w_1 x_{k-1}) \\ \zeta &= \mathbb{E}[\varepsilon_k^2] = \mathbb{E}[(d_k - (x_k + w_1 x_{k-1}))^2] \\ \dots &= \mathbb{E}[d_k^2] - 2w_1 \mathbb{E}[d_k x_k] + w_1^2 \mathbb{E}[x_k^2] + 2w_1 \mathbb{E}[d_k x_{k-1}] - 2w_1^2 \mathbb{E}[x_k x_{k-1}] + w_1^2 \mathbb{E}[x_{k-1}^2]\end{aligned}$$

$$\begin{aligned}\dots \mathbb{E}[d_k x_k] &= -1; \mathbb{E}[x_k x_{k-1}] = 0.5 \\ \implies \zeta &= 7 - w + w^2 = 6.75 + (w - 0.5)^2\end{aligned}$$

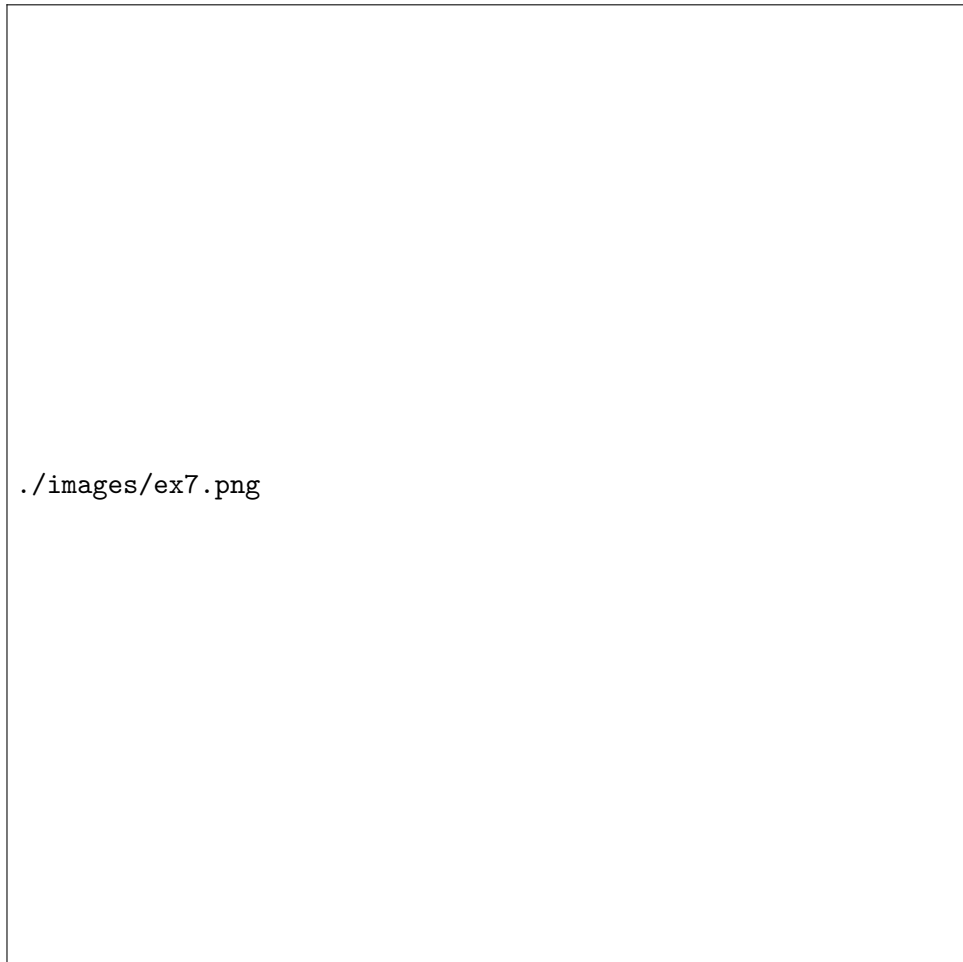


Figure 3: performance function plotted for ex.7

Exercise 8

The optimal value is calculated using the square completion form of the loss function that yields $w_1^* = 1$ with a loss of $\zeta(w_1^*) = r_{min} = 3$

prep exercises set 2 : Searching the Performance Surface

a) Newtons method and SD method

Newton's method is a method for finding a critical point of a function using an iterative algorithm. it updates the input of the function using the gradient and the hessian of the function.

the constraints on Newtons method stability :

In order for the method to work and converge, several conditions must be met:

The function is differentiable

The derivative of the function is continuous

The derivative of the function satisfies the Lipschitz condition

There are additional conditions in cases where there are critical points, additional minimums. In our case, we are dealing with a convex function.

c) SD method

the SD method is similar to Newton's but in contrast it does not use the hessian and performs iterative steps proportionally to the gradient with a ratio μ .

the constraints on the SD method stability is :

$$0 < \mu < \frac{1}{\lambda_{max}}$$

where λ_{max} is the biggest eigenvalue of the correlation matrix of the input x. this condition also promises the convergence of the algorithm to the optimal solution.

The main idea of the method is taking advantage of the fact that the performance function has only one global minimum. If it converges it must be to the optimal solution.

d) proving convergence of SD algo.

We will derive the expression for the weight at an iterative k :

$$\begin{aligned}
 w_k &= w_{k-1} - \mu \nabla_w \zeta(w_{k-1}) ; \nabla_w \zeta(w_k) = 2\lambda(w_k - w^*) \\
 w_k &= w_{k-1} + 2\mu\lambda (w_{k-1} - w^*) = (1 - 2\lambda\mu)w_{k-1} + 2\mu\lambda w^* \\
 &\dots = (1 - 2\lambda\mu)[w_{k-2} - (\nabla_w \zeta(w_{k-2}))] + 2\mu\lambda w^* \\
 &\dots = w_k = (1 - 2\lambda\mu)^k w_0 + \sum_{n=0}^k (1 - 2\lambda\mu) w^* \\
 w_k &= (1 - 2\lambda\mu)^k (w_0 - w^*) + w^*
 \end{aligned}$$

we notice that this expression converges at the limit of $k \rightarrow \infty$ only when the exponent term $1 - 2\mu\lambda$ is smaller than 1 in magnitude.

therefore by enforcing this we get :

$$0 < \mu < \frac{1}{\lambda}$$

which is both the condition for stability and for convergence.

d) comparison between Newtons method and SD

the main difference between Newtons method and SD is that in Newtons method the next iteration step depends on the intersection between the linear approximation of the function with the x coordinate (or the multidimensional equivalent) and the SD method next iteration step depends on the hyperparameter μ .

e) visual comparison between methods

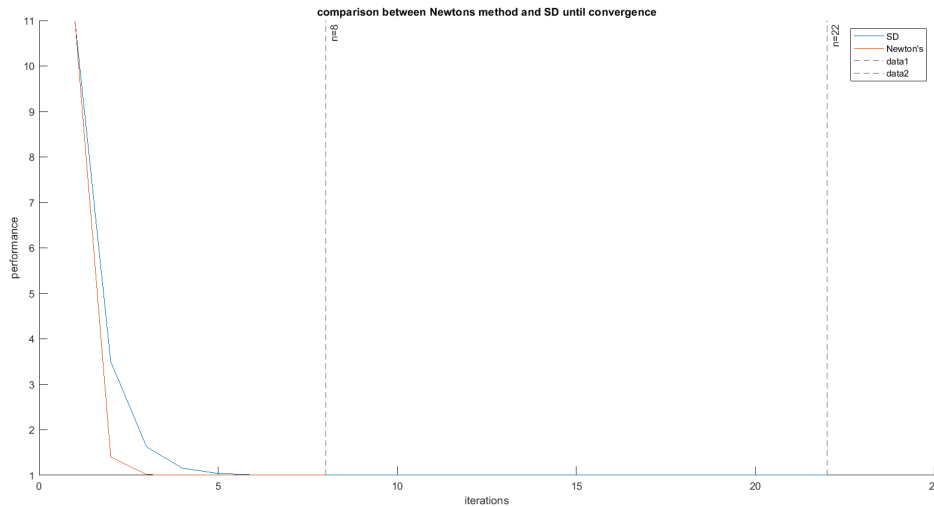


Figure 4: comparison between Newtons method and SD until convergence

we can see that newtons method is more steep than SD and converges. That stems from the reason that newtons method also uses information from the hessian of the function that helps it move towards the direction of the minimum instead of the steepest direction.

f) book exercises 4,5,7,10

ex.4)

The performance function of can be displayed as :

$$\xi = 0.4(w + 5)^2 + 1$$

We can see that $\lambda = 0.4$

since the condition for overdamped weight adjustment is $0 < \mu < \frac{1}{2\lambda}$
we can deduce the condition : $0 < \mu < \frac{1}{0.8} = 1.25$

ex.5)

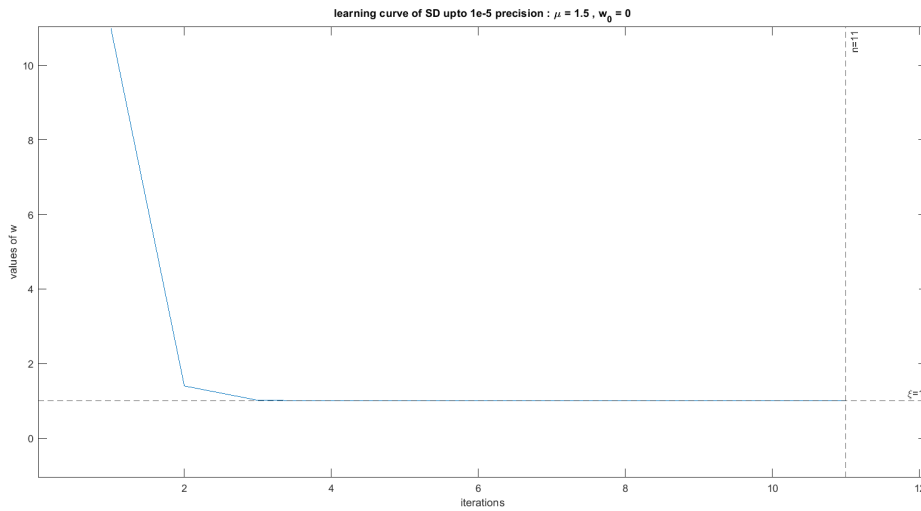


Figure 5: learning curve of SD ($\mu = 1.5, w_0 = 0$) until $1e-5$ precision

ex.7)

- In Figure 4.5, we are given the formula:

$$\xi = 1 - \frac{1}{26} [(1 - w^2)(4 + 3w)^2 + 1]$$

- We derive the first and second derivatives of the function as follows:

$$\frac{d\xi}{dw} = \frac{18w^3 + 36w^2 + 7w - 12}{13}$$

$$\frac{d^2\xi}{dw^2} = \frac{54w^2 + 72w + 7}{13}$$

- And finally, we substitute it into formula 4.26 in the book for the iter. equation and get :

$$w_{k+1} = w_k - \frac{18w^3 + 36w^2 + 7w - 12}{54w^2 + 72w + 7}$$

ex.10)

- we are required to write an algorithm of the form

$$\mathbf{W}_{k+1} = \mathbf{W}_k - \frac{1}{2} \mathbf{R}^{-1} \nabla_k : \forall k \in \mathbb{N} : \nabla_k = 2\mathbf{R}\mathbf{W}_k - 2\mathbf{P}$$

- We use equation 4.29 in the book accordingly

$$\begin{aligned} \xi &= 42 + \begin{bmatrix} w_0 & w_1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} w_0 \\ w_1 \end{bmatrix} - 2 \begin{bmatrix} 7 & 8 \end{bmatrix} \begin{bmatrix} w_0 \\ w_1 \end{bmatrix} \\ &= 2w_0^2 + 2w_1^2 + 2w_0w_1 - 14w_0 - 16w_1 + 42 \end{aligned}$$

- We extract the matrices from the equation

$$\mathbf{R} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \quad \mathbf{P} = \begin{bmatrix} 8 \\ 7 \end{bmatrix} \quad \mathbf{W} = \begin{bmatrix} w_0 \\ w_1 \end{bmatrix}$$

- We calculate $\mathbf{R}^{-1}, \nabla$

$$\begin{aligned} \mathbf{R}^{-1} &= \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \\ \nabla &= 2 \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} w_0 \\ w_1 \end{bmatrix} - 2 \begin{bmatrix} 8 \\ 7 \end{bmatrix} \end{aligned}$$

- We finally get

$$\nabla = \begin{bmatrix} 4w_0 + 2w_1 - 14 \\ 2w_0 + 4w_1 - 16 \end{bmatrix}$$

- When we substitute the expression we derived for the gradient back to the iterative equation, we can see that the algorithm converges to the optimal solution after only one iteration

$$\begin{aligned} \mathbf{W}_{k+1} &= \mathbf{W}_k - \frac{1}{2} \mathbf{R}^{-1} (2\mathbf{R}\mathbf{W}_k - 2\mathbf{P}) \\ &= \mathbf{W}_k - \mathbf{R}^{-1} \mathbf{R}\mathbf{W}_k + \mathbf{R}^{-1} \mathbf{P} \\ &= \mathbf{R}^{-1} \mathbf{P} = \mathbf{W}^* \end{aligned}$$

g) learning curve for formula 3.44

- To plot the learning curve of Newton's method we will use Newton's iteration

$$\xi_k = \xi_{min} + (1 - 2\mu)^{2k} \mathbf{V}_0^T \mathbf{R} \mathbf{V}_0 : \mathbf{V}_0 := (\mathbf{W} - \mathbf{W}_0)$$

- From the original performance function 3.44 we can calculate the value of ξ_{min}

$$\xi_{min} = \xi(2, 3) = 2 \cdot 2^2 + 2 \cdot 3^2 + 2 \cdot 2 \cdot 3 - 14 \cdot 2 - 16 \cdot 3 + 42 = 4$$

- we can derive the performance function of Newton's method to be

$$\xi_k = 4 + (1 - 2 \cdot 0.05)^{2k} ([0 \ 0] - [2 \ 3]) \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix} - \begin{bmatrix} 2 \\ 3 \end{bmatrix} \right)$$

$$\xi_k = 4 + (1 - 2 \cdot 0.05)^{2k} \cdot 38$$

- For the SD method we extract the performance function based on equation 4.59 from the book where ξ_{min} was calculated previously

$$\xi_k = \xi_{min} + \sum_{n=0}^{\infty} v'_{on}{}^2 \lambda_n (1 - 2\mu)^{2k}$$

- we perform an SVD to the matrix $\mathbf{R}$ to derive this representation

$$\Lambda = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}, \quad Q = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix} = Q^{-1}$$

$$\mathbf{R} = \mathbf{Q} \Lambda \mathbf{Q}^{-1}$$

$$\mathbf{V}' = -\frac{1}{\sqrt{2}} \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 3 \end{bmatrix} = -\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 5 \end{bmatrix}$$

- we can substitute the matrix $\mathbf{V}'$ into the function earlier and get the final performance equation for the SD method

$$\xi_k = 4 + \left(-\frac{1}{\sqrt{2}} \right)^2 (1 - 2 \cdot 0.05 \cdot 1)^{2k} + \left(-\frac{5}{\sqrt{2}} \right)^2 (1 - 2 \cdot 0.05 \cdot 3)^{2k}$$

$$\xi_k = 4 + 0.5(0.9)^{2k} + 12.5(0.7)^{2k}$$

- we plot the 2 functions together for comparison. we can see that SD outperforms Newton's method.

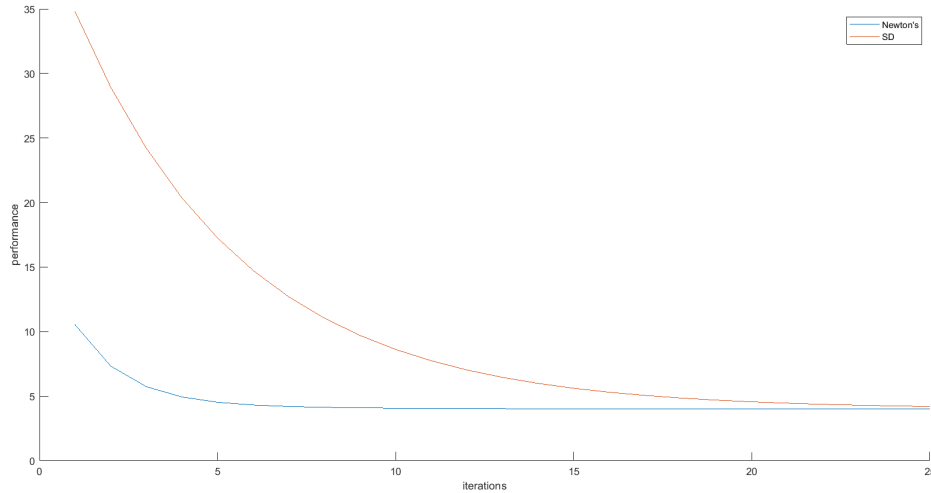


Figure 6: learning curve of Newton and SD ($\mu = 1.5, w_0 = 0$) until $1e-5$ precision

prep exercises set 3 : Gradient estimation and effects on adaptation

a) explaining term set

1. Performance Penalty

This represents the cost of using an adaptive algorithm rather than an optimal fixed filter. It accounts for the fact that a system must "learn" and track changes, meaning it can never stay perfectly at the theoretical minimum error point, resulting in a higher error than the Wiener solution.

2. Perturbation

Perturbation refers to the random fluctuations in filter weights caused by gradient estimation noise. Because adaptive filters use stochastic data to update coefficients, the weights "chatter" around the optimal values, preventing the system from reaching a perfectly static state.

3. Excess Mean-Square Error (EMSE)

EMSE is the difference between the actual Mean-Square Error (MSE) achieved by the adaptive filter and the minimum possible MSE (J_{min}). *Importance:* It is a critical metric because it quantifies the specific amount of additional noise introduced into the signal solely due to the adaptive process, allowing engineers to measure the "efficiency" of the algorithm.

4. Misadjustment (M)

Misadjustment is a dimensionless ratio defined as the EMSE divided by the Minimum MSE ($M = EMSE/J_{min}$). It provides a normalized measure of performance, allowing for a fair comparison between different systems regardless of the input signal power.

5. Total Misadjustment

In complex or multi-stage systems, this is the cumulative sum of all misadjustments across the processing chain. *Importance:* It provides the "bottom-line" figure for total system degradation, which is essential for determining if the aggregate error remains within acceptable limits for applications like high-speed telecommunications or echo cancellation.

b) book exercises 2,4,6

ex.2) illustrating performance penalty of an adaptive system

- We are given a performance surface of an adaptive system

$$\xi = 5w^2 - 20w + 23$$

- With the following display of the function we can see the parameters of the function

$$\xi = 5(w - 2)^2 + 3 \implies \{\lambda = 5, \quad \xi_{min} = 3, \quad w^* = 2\}$$

- The performance penalty of perturbing around $w = 2.5$ with perturbation of $\pm\delta = \pm 1$ is given by the following expression

$$\gamma = \lambda\delta^2, \quad \pm\delta = \pm 1$$

$$\gamma = 5 \cdot 1^2 = 5$$

- And we get that the total penalty of perturbing around $w = 2.5$ is given by

$$P = \frac{\gamma}{\xi_{min}} = \frac{5}{3}$$

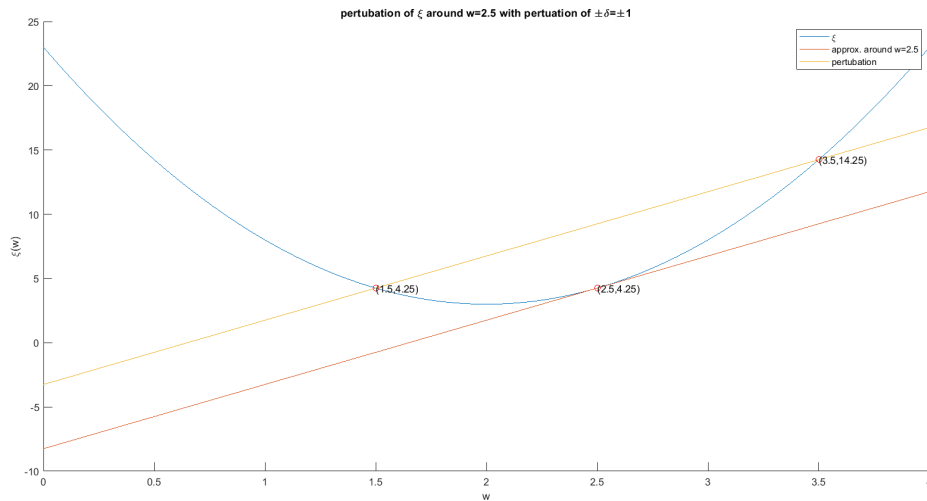


Figure 7: perturbation in performance around $w = 2.5$ with perturbation of $\pm\delta = \pm 1$

ex.4) * value of perturbation calculated previously, see ex.2

ex.6) weight addition

Suppose that we add a weight to a given transversal filter. Since the perturbation is linear with the trace of the covariance matrix, adding another weight adds a value σ_{new}^2 to that trace (which is positive) and by that increasing the perturbation.

prep exercises set 4 : LMS algorithm

a) Motivation

The motivation for using the LMS algorithm stems from 2 core principals :

1. Low computational complexity - $\mathcal{O}(L^2)$ where L is the number of tap inputs of the system. considered light compared to other methods.
2. Simplicity - requires no No prior data or statistical knowledge of the distribution of noise or any effect and instead estimates the gradient instantaneously.

b) Gradient estimation

- We use the error and input to estimate the gradient

$$\nabla = -2(d_k - y_k) \cdot \mathbf{x}_k = -2\varepsilon_k \cdot \mathbf{x}_k$$

- we substitute back to the SD iter. equation

$$\mathbf{W}_{k+1} = \mathbf{W}_k - \mu \nabla_k = \mathbf{W}_k - \mu 2\varepsilon_k \cdot \mathbf{X}_k$$

c) Unbiased estimator - We will show that the LMS gradient estimation is unbiased

$$\begin{aligned}\text{Bias} &= \mathbb{E}[\widehat{\nabla}] - \nabla \\ \text{Bias} &= \mathbb{E}[-2 \cdot \varepsilon \cdot X] - 2(RW - P) \\ \text{Bias} &= \mathbb{E}[-2 \cdot (d - y) \cdot X] - 2(RW - P) \\ \text{Bias} &= -2 \cdot (\mathbb{E}[dX] - \mathbb{E}[XX^T]W) - 2(RW - P) \\ \text{Bias} &= -2 \cdot (P - RW) - 2(RW - P) = 0\end{aligned}$$

Since the expected value of estimation error is 0, we conclude that the estimator is unbiased.

d) Algorithm derivation

The equation previously derived is the algorithm when written on code. see derivation in 4.a)

e) Convergence to optimal solution

We will derive the expression of the expected value of the weight vector to show that it converges to the optimal solution in the limit as the iteration number goes to infinity

- taking the expected value on both sides of the iter. equation

$$\begin{aligned} W_{k+1} &= W_k - 2\mu\varepsilon_k X_k \\ \mathbb{E}[W_{k+1}] &= \mathbb{E}[W_k + 2\mu\varepsilon_k X_k] \\ \mathbb{E}[W_{k+1}] &= \mathbb{E}[W_k] + 2\mu\mathbb{E}[dX - XX^T W_k] \end{aligned}$$

- assuming $\mathbf{X} \perp \mathbf{W}$ we get

$$\mathbb{E}[W_{k+1}] = \mathbb{E}[W_k] + 2\mu(\mathbb{E}[dX] - \mathbb{E}[XX^T]\mathbb{E}[W_k])$$

- substituting $\mathbf{R}, \mathbf{P}$ as defined we get

$$\begin{aligned} \mathbb{E}[W_{k+1}] &= \mathbb{E}[W_k] + 2\mu(P - R\mathbb{E}[W_k]) \\ \mathbb{E}[W_{k+1}] &= \mathbb{E}[W_k] - 2\mu R\mathbb{E}[W_k] + 2\mu P \\ \mathbb{E}[W_{k+1}] &= \mathbb{E}[W_k](I - 2\mu R) + 2\mu P \end{aligned}$$

- We have already showed for our case that $\mathbf{W}^* = \mathbf{R}^{-1}\mathbf{P}$

$$\mathbb{E}[W_{k+1}] = \mathbb{E}[W_k](I - 2\mu R) + 2\mu RW^*$$

- we define $V'_k := W_k - W^*$

$$\mathbb{E}[V'_k] = (I - 2\mu\Lambda)^k \cdot V'_0$$

- assuming that all eigenvalues are < 1 we get

$$\begin{aligned} \lim_{k \rightarrow \infty} \mathbb{E}[V'_k] &= 0 \\ \lim_{k \rightarrow \infty} \mathbb{E}[W_k - W^*] &= 0 \\ \lim_{k \rightarrow \infty} \mathbb{E}[W_k] &= W^* \end{aligned}$$

We have shown that the estimator converges to the optimal solution in the sense of MMSE.

f) Condition of convergence

- We assumed in previous subsection that all eigenvalues are smaller than 1 which means that the condition is as follows

$$0 < \mu < \frac{1}{\lambda_{max}}$$

- Since knowing the max eigenvalue requires decomposition of a matrix of L by L the complexity of it is $\mathcal{O}(L^3)$ which is unnecessarily heavy. We can calculate the Trace of the correlation matrix of the tap inputs since it is the sum of its eigenvalues. That is justified under the assumptions that the tap inputs are wide-sense stationary (WSS).

$$\lambda_{max} \leq \text{tr}(\Lambda) = \text{tr}(R) = (L+1)\mathbb{E}[x_k^2] = (L+1) \cdot \sigma_x^2$$

$$0 < \mu < \frac{1}{\text{tr}(R)} \leq \frac{1}{\lambda_{max}}$$

- Calculation of the trace is $\mathcal{O}(L)$ which is tremendously lighter in computation. This is compelling/forcing condition but not a necessary one.

g) book exercises 2,7,8

ex.2) adaptive gain constant

we have previously shown that the condition of convergence under the specified given system is

$$0 < \mu < \frac{1}{\text{tr}(R_x)} = \frac{1}{(L+1)p} \leq \frac{1}{\lambda_{max}}$$

see derivation in 4.f)

ex.7)

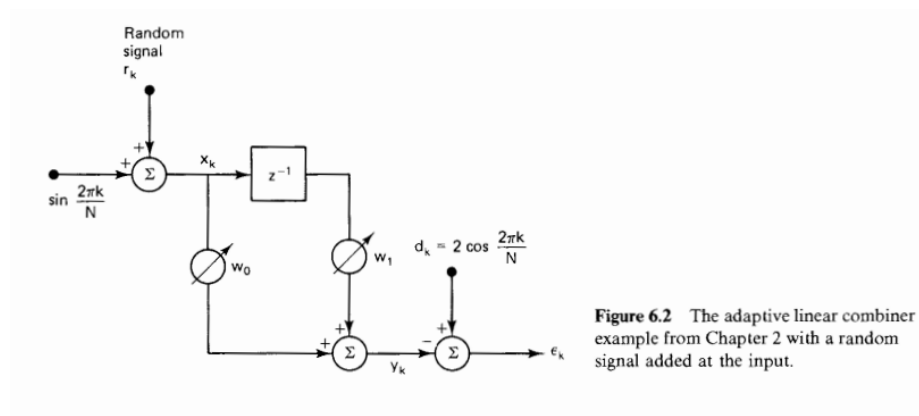


Figure 8: 6.2 - block diagram from book

- a) if we add another weight the algorithm will use a broader range of the signal in time domain and use more information about it and that will diminish the MSE.

From this result we may compute the misadjustment, defined above and in (5.96), as

$$\boxed{\begin{aligned} M &= \frac{\text{excess MSE}}{\xi_{\min}} \\ &\approx \mu \text{tr}[\mathbf{R}] \end{aligned}} \quad (6.36)$$

Figure 9: 6.36 - Misadjustment expression for LMS algorithm from book

- b) We can see that adding another weight and tap input will increase the trace of the matrix $\mathbf{R}$ and therefore increase the misadjustment.

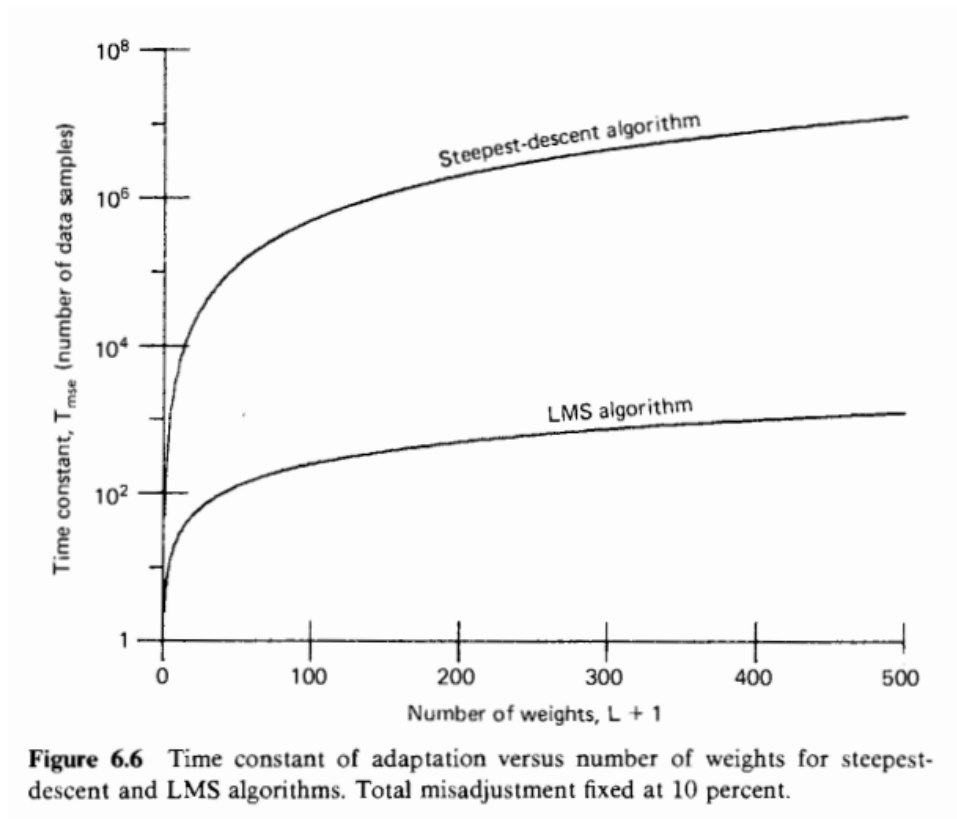


Figure 10: 6.36 - Misadjustment expression for LMS algorithm from book

- c) we can see from the behavior of the time constant that an increase in in the number of tap inputs increases the time constant resulting in a less steep learning curve.

ex.8)

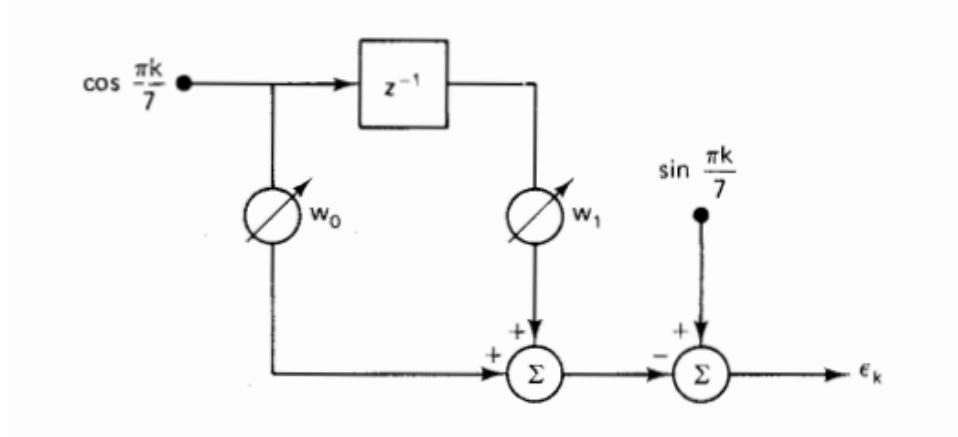


Figure 11: block diagram exercise 8

- **Step 1: Define the input vector.**

The input vector X is composed of the current sample and the immediately preceding sample of a cosine signal:

$$X = \begin{pmatrix} \cos\left(\frac{\pi k}{7}\right) & \cos\left(\frac{\pi(k-1)}{7}\right) \end{pmatrix}$$

- **Step 2: Define the weight vector.**

The corresponding weight vector W for a two-tap adaptive filter is defined as:

$$W = \begin{pmatrix} w_0 & w_1 \end{pmatrix}$$

- **Step 3: Compute the Autocorrelation Matrix.**

The autocorrelation matrix R is found by taking the expectation of the outer product of the input vector X with its transpose X^T :

$$R = \mathbb{E}[XX^T] = \mathbb{E} \begin{bmatrix} \cos^2\left(\frac{\pi k}{7}\right) & \cos\left(\frac{\pi k}{7}\right) \cos\left(\frac{\pi(k-1)}{7}\right) \\ \cos\left(\frac{\pi k}{7}\right) \cos\left(\frac{\pi(k-1)}{7}\right) & \cos^2\left(\frac{\pi(k-1)}{7}\right) \end{bmatrix}$$

- **Step 4: Calculate the diagonal elements (Signal Power).**

The expectation of the squared cosine terms represents the average power of the signal:

$$\mathbb{E} \left[\cos^2\left(\frac{\pi k}{7}\right) \right] = \mathbb{E} \left[\cos^2\left(\frac{\pi(k-1)}{7}\right) \right] = \frac{1}{7} \sum_{k=0}^6 \cos^2\left(\frac{\pi k}{7}\right) = \frac{1}{2}$$

- **Step 5: Calculate the off-diagonal elements (Cross-correlation).**

Using the product-to-sum trigonometric identity, the expectation of the product of the two taps is computed as:

$$\begin{aligned} \mathbb{E} \left[\cos\left(\frac{\pi k}{7}\right) \cos\left(\frac{\pi(k-1)}{7}\right) \right] &= \frac{1}{7} \sum_{k=0}^6 \cos\left(\frac{\pi k}{7}\right) \cos\left(\frac{\pi(k-1)}{7}\right) \\ &= \frac{1}{2} \cdot \frac{1}{7} \sum_{k=0}^6 \left[\cos\left(\frac{2\pi k - \pi}{7}\right) + \cos\left(\frac{\pi}{7}\right) \right] \end{aligned}$$

- **Step 6: Simplify the cross-correlation result.**

Since the sum of a cosine over a full period is zero, the term involving k vanishes, leaving:

$$= \frac{1}{2} \cos\left(\frac{\pi}{7}\right) + \frac{1}{2} \cdot \frac{1}{7} \sum_{k=0}^6 \cos\left(\frac{2\pi k - \pi}{7}\right) = \frac{1}{2} \cos\left(\frac{\pi}{7}\right)$$